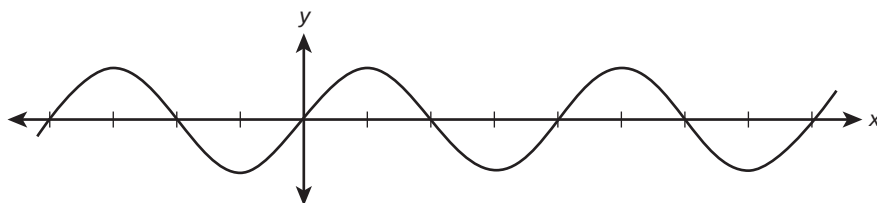


Sinusoidal Graphs

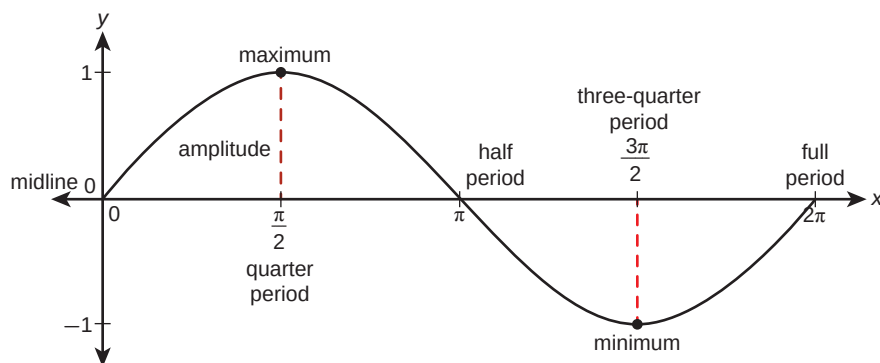
Period, midline, and amplitude are some of the key characteristics of sinusoidal functions.

Identifying Characteristics of Sinusoidal Graphs

Any sine or cosine function has a graph that resembles the one shown. These functions are called **sinusoidal functions**.



Here is sinusoidal $f(x)$ graphed on the interval $(0, 2\pi)$.



Notice that the maximum height the function reaches is 1. So the maximum is 1. The function's minimum height is -1 , so the minimum is -1 .

- **Remember** Recall that the period of a function is the smallest interval where the function repeats itself. In the case of $f(x)$, the period is 2π .

Definitions

The **midline** of a sinusoidal function is the horizontal line that is halfway between the maximum and the minimum. The equation of the midline of a sinusoidal function is

$$y = \frac{\text{maximum} + \text{minimum}}{2}.$$

The **amplitude** of a sinusoidal function is one-half the distance between the maximum and minimum values.

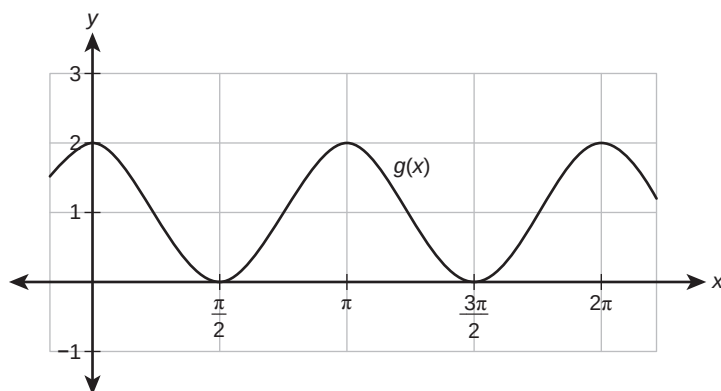
$$\text{amplitude} = \frac{\text{maximum} - \text{minimum}}{2}$$

- **Remember** If the minimum y-value is -1 and the maximum y-value is 1 , then the range of the function is $[-1, 1]$.

EXAMPLE

Determine the period, minimum, maximum, midline, and amplitude of the function.

A



SOLUTION

Since the graph of $g(x)$ repeats itself every π , the period is π . The lowest y -value the function takes on is 0, so the minimum is 0. The maximum value the function reaches is 2, so the maximum is 2.

$$\text{equation of midline: } y = \frac{\text{maximum} + \text{minimum}}{2} = \frac{2 + 0}{2} = 1$$

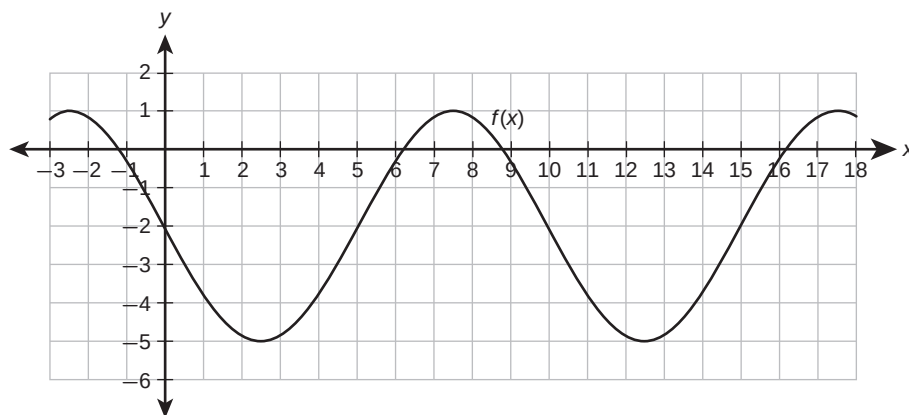
$$\text{amplitude} = \frac{\text{maximum} - \text{minimum}}{2} = \frac{2 - 0}{2} = 1$$

► Q&A

Q What is the range of $g(x)$ from Example A?

A $[0, 2]$

B



SOLUTION

The period is 10 since the function repeats every 10. The minimum is -5 , and the maximum is 1.

$$\text{equation of midline: } y = \frac{1 + (-5)}{2} = -2$$

$$\text{amplitude} = \frac{1 - (-5)}{2} = 3 \quad \blacksquare$$